

EDA Bench: Execution-Based Evaluation for PCB Agents

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FAGEN workshop, ICML 2026

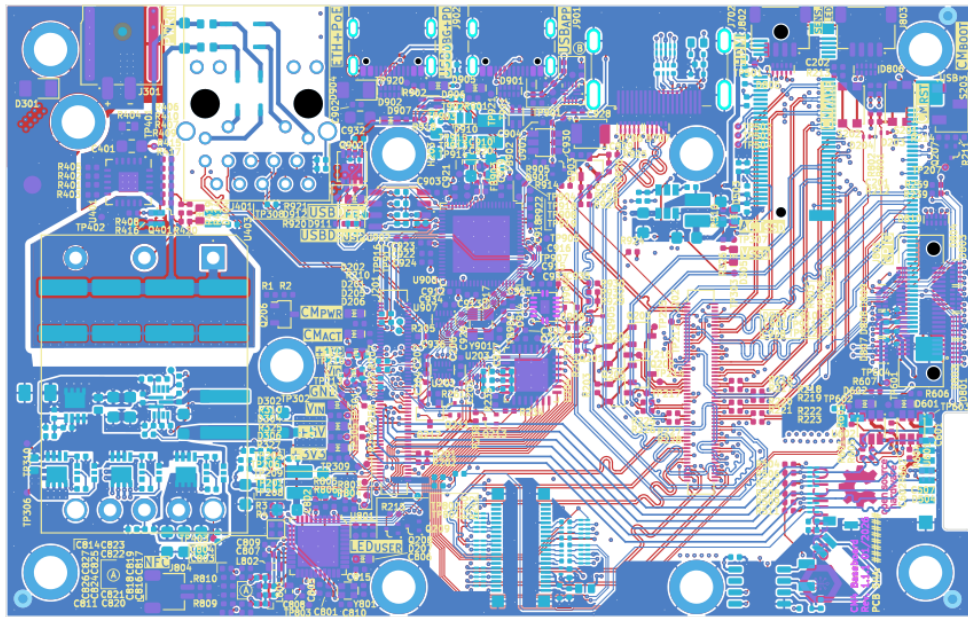
Motivation

Language-model agents can search for parts, write board files, run design tools, and repair parser errors. Weak evaluation signals over-credit that progress:

- a board-design file exists;
- the file opens in the design tool;
- components appear in roughly plausible places;
- a screenshot looks like a circuit board.

Those proxies miss the engineering question: whether the generated board preserves the requested external electrical behavior at the connectors and pins a downstream user would touch.

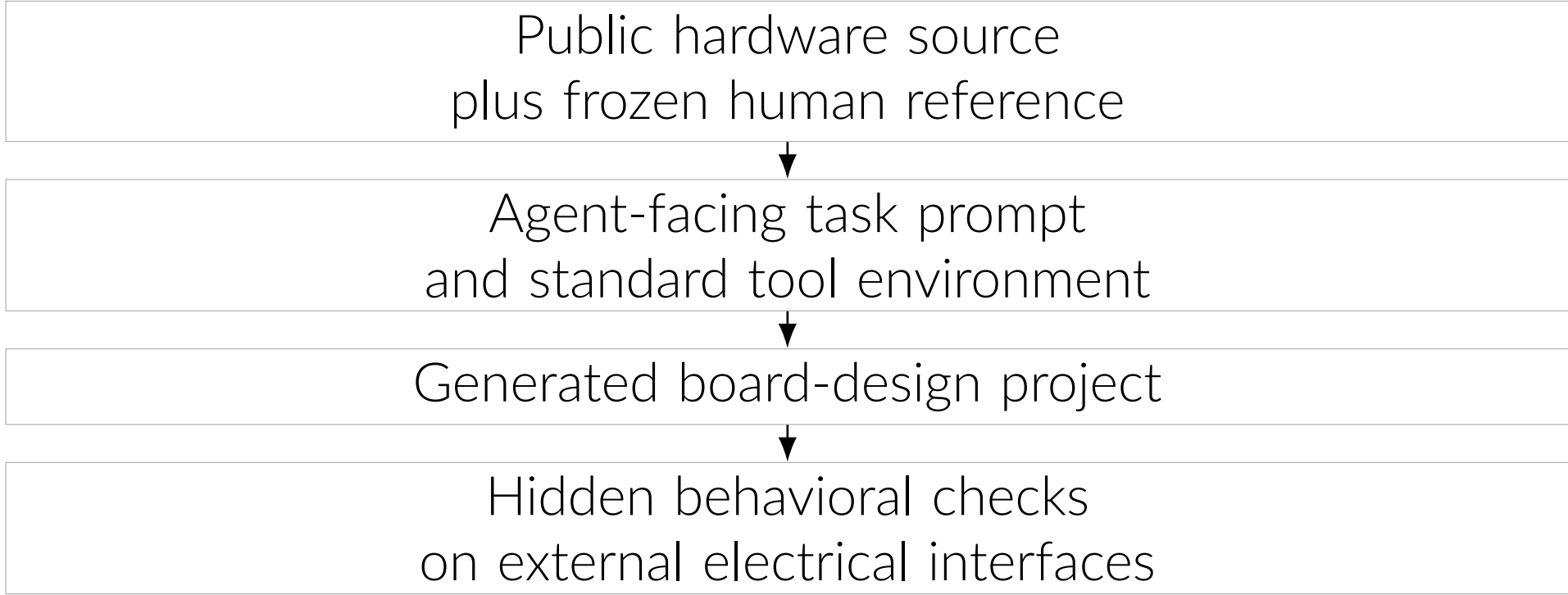
Task Artifact



Task set	35 benchmark tasks from open-hardware projects
Outputs	complete circuit-board design projects produced during evaluation
Scoring	external I/O behavior; routed copper; shorts; powered circuitry; design rule diagnostics
Proxies	screenshots, labels, component overlap

CM4 carrier target; scoring uses generated connector and pin behavior.

Construction

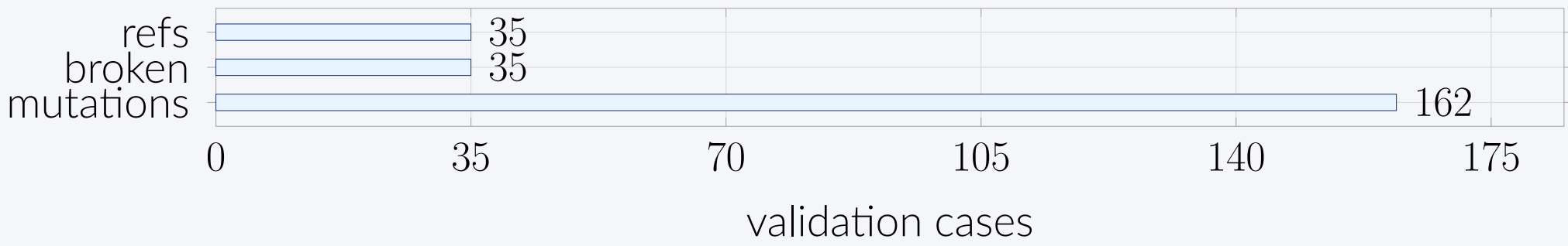


Tasks use open-hardware projects, frozen reference designs, and hidden interface contracts. Agents submit generated board projects for behavioral scoring.

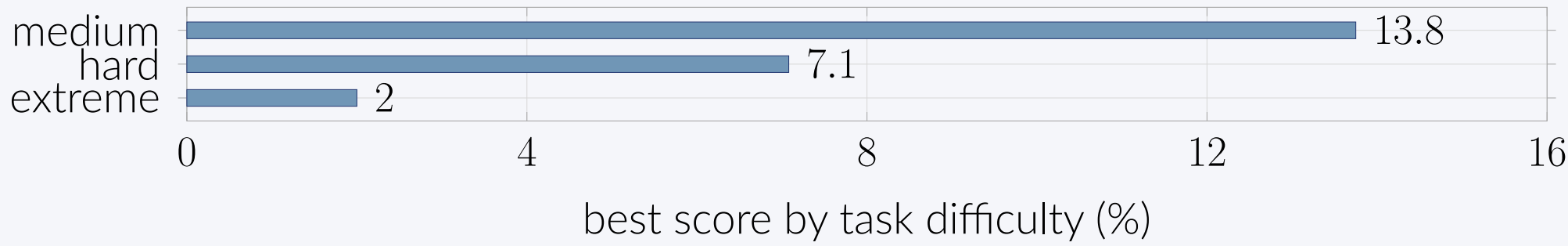
Scoring Signals

Ports	connector pads bind to task-declared external interfaces
Routing	continuity, isolated pads, shorts, and trace geometry are checked from the board
Behavior	probes score external I/O response and active circuitry
Tool checks	ERC and DRC diagnostics contribute but are not the whole metric

Controls and Difficulty

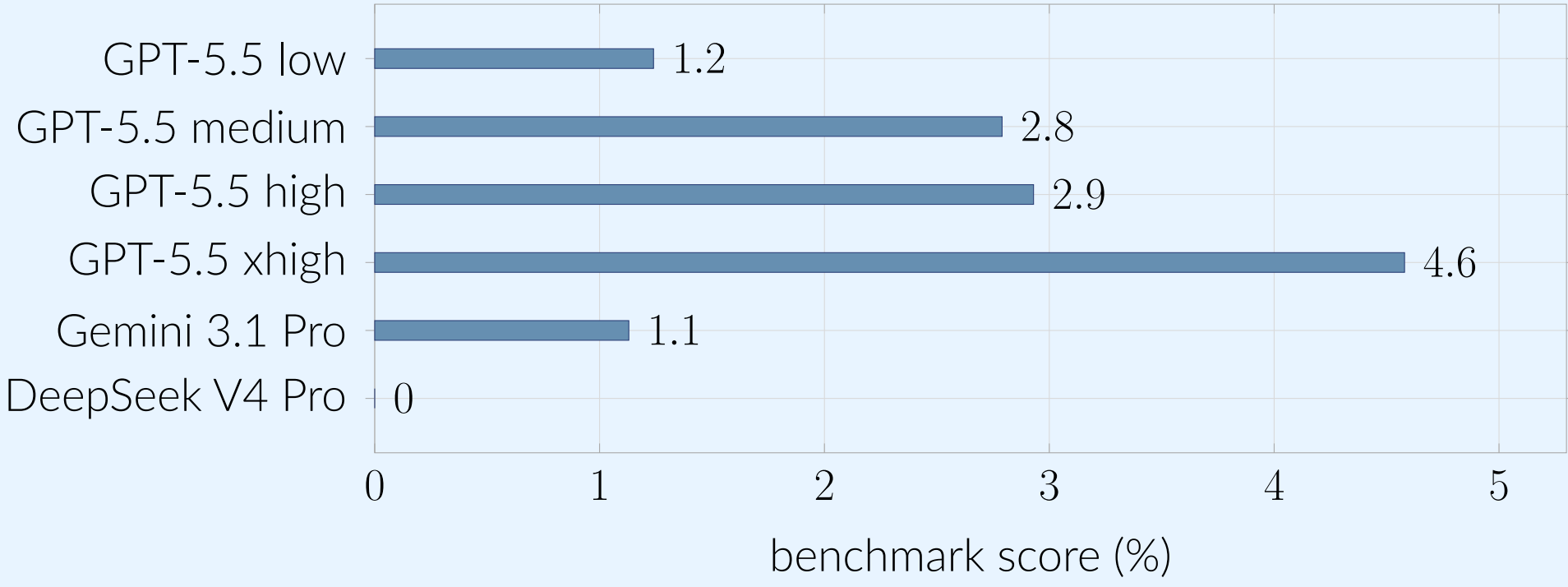


References score 1.0; broken shells and mutations stay below threshold.

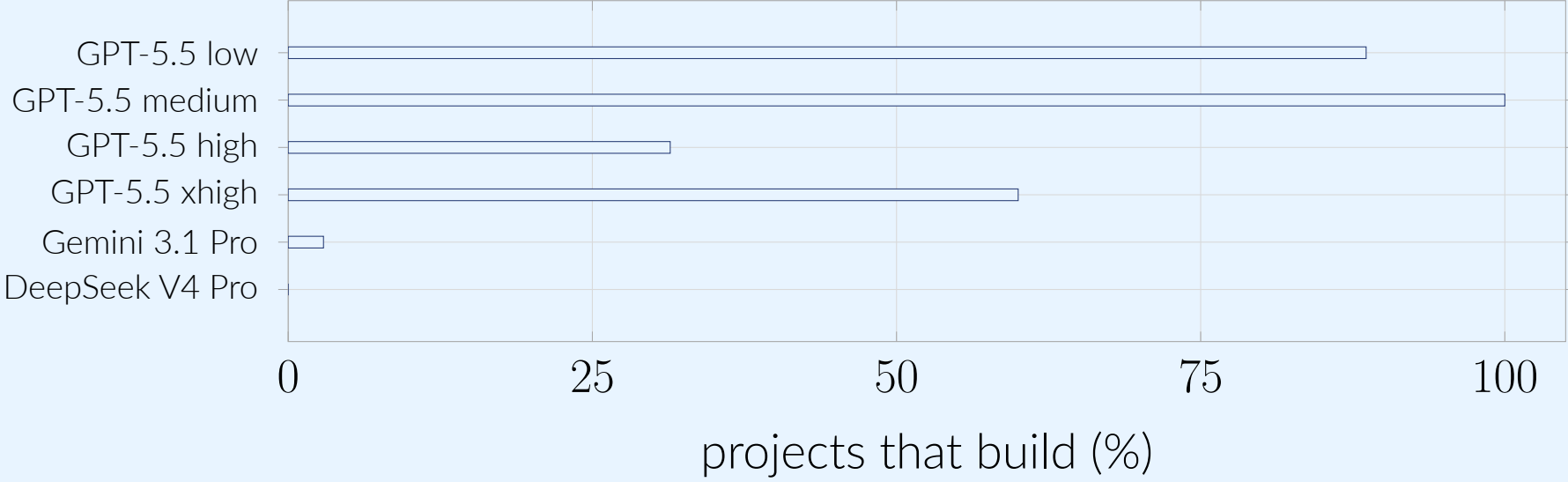


GPT-5.5 xhigh drops from 13.8% on medium tasks to 2.0% on extreme tasks.

Scores and Builds

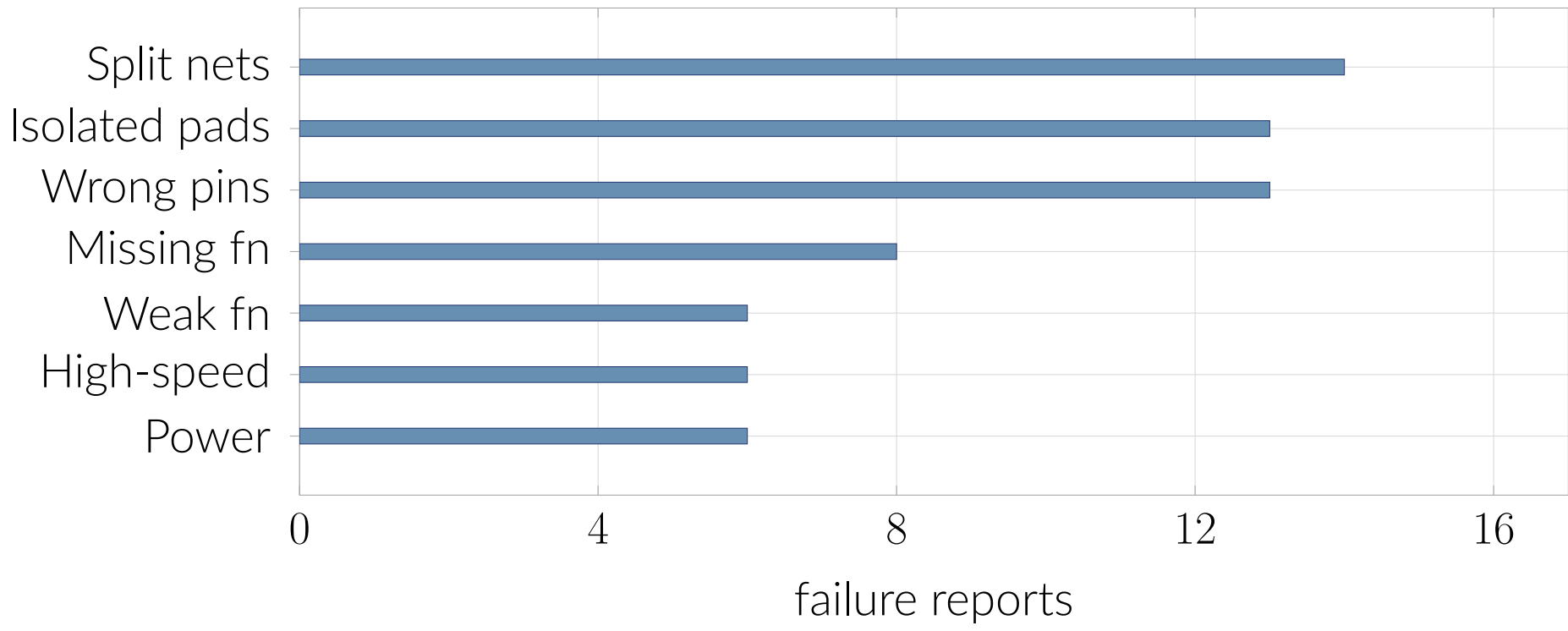


GPT-5.5 is evaluated at low, medium, high, and xhigh reasoning settings. Gemini 3.1 Pro and DeepSeek V4 Pro use high reasoning settings.



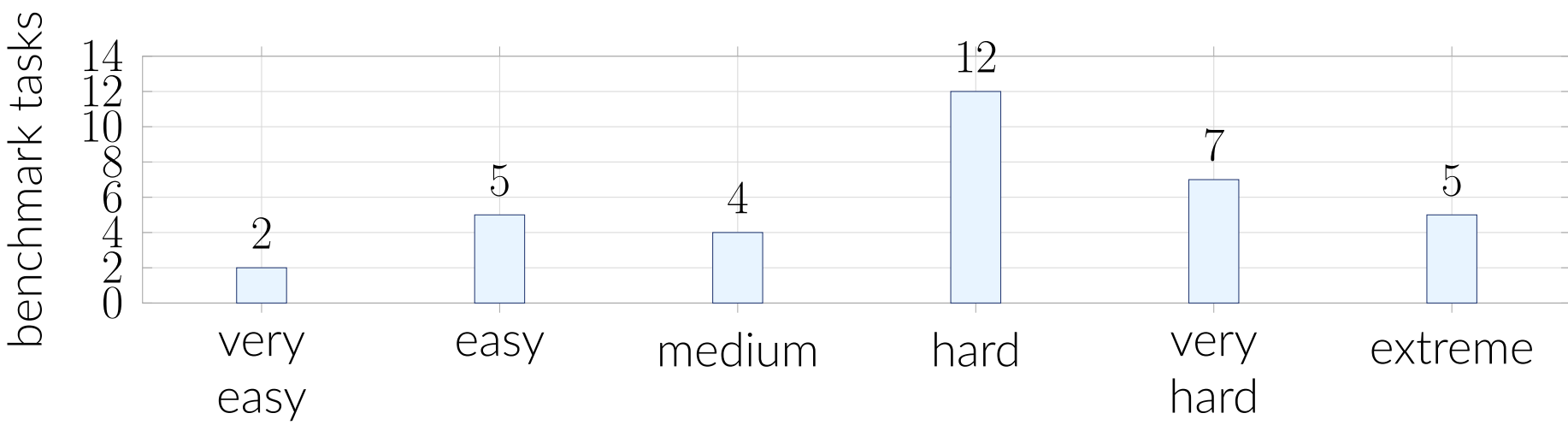
GPT-5.5 medium builds all 35 projects but scores 2.79%; GPT-5.5 xhigh builds 21/35 and scores 4.58%.

Failure Modes



Failures concentrate on continuity, isolated pads, wrong external pin semantics, weak circuitry, and implausible high-speed-like routing.

Task Mix



The catalog spans compact breakouts, power-path circuits, controller boards, compute and camera adapters, motor drivers, and dense carriers.

Key Results

Best score	GPT-5.5 xhigh reaches 4.58%, still far from functional reconstruction.
Build proxy	GPT-5.5 medium builds every project but scores 2.79%.
Main signal	runnable board files and plausible layouts are not enough; external electrical behavior is the target.

Evaluation Scope

Scores measure complete agent systems: model, reasoning setting, browser/tool access, generated project, prompts, and the fixed grader version.